04-DETERMINATION of the e/m RATIO of the ELECTRON

PURPOSE of the EXPERIMENT

Determination of the e/m ratio of the electron by studying the motion of cathode rays accelerated under different electric potentials in a uniform magnetic field.

GENERAL INFORMATION

If a free electron with charge e moves with a velocity v in a vacuum perpendicular to a magnetic field B, the magnetic force acting on this charge is given by the following relation.

$$\overrightarrow{F_B} = e\vec{v}x\vec{B} \qquad (1)$$

In the experimental setup, electrons released by heating the cathode are accelerated under a voltage V and sent between Helmholtz coils. Since there is no magnetic field as long as no current flows through the Helmholtz coils, the electron beam continues on its way without any deflection.

The electrical energy of the electrons with charge e accelerated by the accelerator voltage V is converted into the kinetic energy of the electron eV.

$$eV = \frac{1}{2}mv^2 \qquad (2)$$

This gives the following expression.

$$v = \left(\frac{2eV}{m}\right)^{1/2} \qquad (3)$$

A magnetic field is created between the coils by passing a current between 0-1A through Helmholtz coils. As the current is increased, a uniform magnetic field B is created in the region between the coils. The strength of the magnetic field is set so that it is much larger than the force of gravity so that the force of gravity is ignored. The magnetic force exerted on the electron by the magnetic field is perpendicular to the electron's velocity and forces the electron

into a circular motion. This magnetic force provides the centripetal force necessary to move in a circular orbit.

$$m\frac{v^2}{r} = evB \qquad (4)$$

If v is substituted here and the e/m ratio is created, the following expression is obtained.

$$\frac{e}{m} = \frac{2V}{B^2r^2} \qquad (5)$$

As can be seen from this expression, the ratio of the electron's charge to its mass is determined by the variation of the accelerating voltage concerning the magnetic field for a circular orbit of constant radius.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup is as follows.



Figure 1: Experimental setup

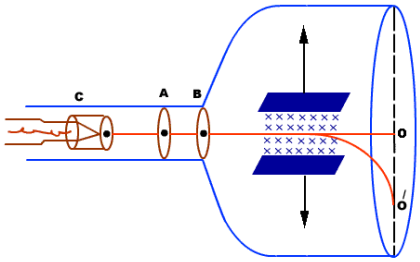
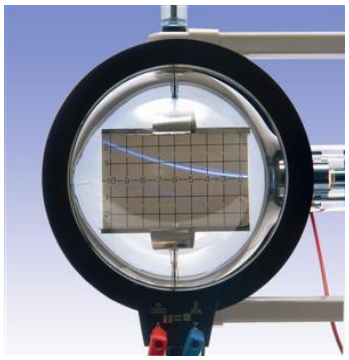
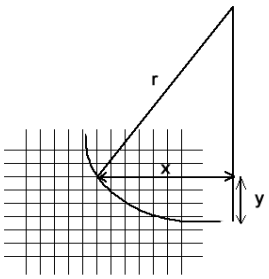


Figure 2: Cathode ray tube

Determination of r

The radius of the circular orbit of the electron beam on the fluorescent screen can be calculated from the figure:





$$r^2 = x^2 + (r - y)^2$$

$$r = (x^2 + y^2)/2y \qquad (6)$$

Figure 3: On-screen image of the electron trajectory.

Determination of B

One method for obtaining a uniform magnetic field is to use a pair of Helmholtz coils with equal currents flowing through them and with a common axis. For a coil of a given radius, the spacing between the coils must be equal to the radius to obtain the optimum uniform field.

The magnetic field between the coils is given by the Biot-Savart law, as follows,

$$B = \frac{\mu_0 N R^2}{(R^2 + a^2)^{3/2}} I \qquad \mu_0 = 1 \qquad (7)$$

N: number of turns on each coil (given as 320)

R: radius of the coil (6.8 cm)

a: average distance between coils (6.8 cm)

I: the amount of current flowing through each coil.

Thus the relation between current and magnetic field is given by $B = 4{,}25.10^{-3}\ m^{-1}\ I$ when the constants are substituted.

As the current increases, the magnetic field strength increases accordingly. In the experiment, the e/m_e ratio is calculated by measuring V, I, x, and y values.

V (Volt)	**I (A)**	**r (cm)**	**B (Tesla)**	***e*/*m*$_e$ (C/kg)**
Average *e*/*m*$_e$ =				

REFERENCES

1. http://hyperphysics.phy-astr.gsu.edu/hbase/magnetic/helmholtz.html
2. Phywe experimental literature and operating manuals